

Spring 2011

Problem 1 (Sp11). 1. Prove that every path connected space is connected.

2. If X is the subset of the plane consisting of the points (x, y) with $x = 0$ or $x > 0, y = \sin(1/x)$ show that X is connected but not path connected.

Note: See also ??.

Problem 2 (Sp85, Sp11, Fa17, Sp21). Find an irreducible polynomial over the integers with $2 \cos(2\pi/7)$ as a zero, and use this to show that it is not contained in any extension of the rational numbers of degree a power of 2.

Problem 3 (Fa77, Su82, Fa97, Sp01, Fa03, Fa08, Sp11, Fa12, Sp18, Sp21).

Evaluate $\int_{-\infty}^{\infty} \frac{dx}{1+x^{2n}}$, where n is a positive integer.

Problem 4 (Sp11). Let $M_{n \times n}(\mathbb{F})$ be the $n \times n$ matrices over a field $\mathbb{F}$. Find all linear maps $f : M_{n \times n}(\mathbb{F}) \rightarrow \mathbb{F}$ such that $f(AB) = f(BA)$ for all matrices A and B .

Problem 5 (Sp12, Sp11). Suppose the power series $\sum_n a_n x^n$ converges for all real x , and the smooth real valued function f has the property that

$$\lim_{x \rightarrow 0} \frac{f(x) - \sum_{j=0}^n a_j x^j}{x^n} = 0$$

for all n . Prove or give a counterexample to the claim that $f(x) = \sum_n a_n x^n$.

Problem 6 (Sp11). If N is the integer $2^4 \cdot 3^3 \cdot 5^2 \cdot 7$ find the smallest positive integer m such that $x^m \equiv 1 \pmod{N}$ for all integers x coprime to N .

Problem 7 (Sp11, Sp21). If $0 < r < 1$, find $\sum_{k=0}^{\infty} r^k \cos k\theta$.

Problem 8 (Sp11, Sp21). For each of the following 4 statements, give either a counterexample or a reason why it is true.

1. For every real matrix A there is a real matrix B with $B^{-1}AB$ diagonal.
2. For every complex matrix A there is a complex matrix B with $B^{-1}AB$ diagonal.

3. For every symmetric complex matrix A there is a complex matrix B with $B^{-1}AB$ diagonal.
4. For every symmetric real matrix A there is a real matrix B with $B^{-1}AB$ diagonal.

Problem 9 (Sp11, Sp26). The Catalan numbers C_n satisfy $C_0 = 1$, $C_n = C_0C_{n-1} + C_1C_{n-2} + \cdots + C_{n-1}C_0$ for $n > 0$. Find the function $\sum_{n=0}^{\infty} C_n x^n$ and use this to evaluate C_n .

Problem 10 (Sp11). Let D be an open subset of $\mathbb{R}^2$, with the topology induced by the Euclidean metric, and assume that it contains the closed unit square

$$[0, 1] \times [0, 1] = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1\}$$

Show that D contains the partially-open rectangle

$$[0, 1] \times [0, 1 + \varepsilon) = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y < 1 + \varepsilon\}$$

for some $\varepsilon > 0$.

Problem 11 (Su80, Fa96, Sp11). Prove that every finite group is isomorphic to

1. A subgroup of a group of permutations of n objects;
2. A subgroup of the group of even permutations of n objects.

Problem 12 (Sp11, Fa12). Show that $e^z = z$ has infinitely many complex roots.

Problem 13 (Sp11). For which real numbers x does the matrix valued series $\sum_{n=0}^{\infty} x^n A^n$ converge, where A is the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$?

Problem 14 (Sp11). 1. Evaluate $I(n) = \int_0^{\pi} \sin^n x \, dx$ for n a nonnegative integer.

2. Prove that $I(n) > I(n+1) > 0$.

3. Evaluate the infinite product $\frac{1}{2} \cdot \frac{3}{2} \cdot \frac{3}{4} \cdot \frac{5}{4} \cdot \frac{5}{6} \cdots$

Problem 15 (Sp82, Sp11, Fa17, Sp21). Prove that the polynomial $x^4 + x + 1$ is irreducible over $\mathbb{Q}$.

Problem 16 (Sp11). Prove that the real and imaginary parts of a holomorphic complex function are harmonic. Find two linearly independent real solutions of Laplace's equation in two variables that are homogeneous polynomials of degree 6.

Problem 17 (Su79, Sp82, Sp83, Su84, Fa91, Fa98, Sp11, Fa15, Fa16, Fa23). Let $\mathbb{F}$ be a finite field with q elements and let V be an n -dimensional vector space over $\mathbb{F}$.

1. Determine the number of elements in V .
2. Let $\text{GL}_n(\mathbb{F})$ denote the group of all $n \times n$ nonsingular matrices over $\mathbb{F}$. Determine the order of $\text{GL}_n(\mathbb{F})$.
3. Let $\text{SL}_n(\mathbb{F})$ denote the subgroup of $\text{GL}_n(\mathbb{F})$ consisting of matrices with determinant 1. Find the order of $\text{SL}_n(\mathbb{F})$.

Problem 18 (Sp82, Fa08, Sp11, Sp25). For each of the following statements, either prove it or give a counterexample:

1. If $f_n(x)$ are continuous real valued functions on the unit interval, and f_n converges pointwise to the function f , that is, $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ for all x , then

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) \, dx = \int_0^1 f(x) \, dx$$

2. If f_n are continuous and converges uniformly to a function f then the sequence $\int_0^1 f_n(x) \, dx$ has limit $\int_0^1 f(x) \, dx$.
3. If $g(m, n)$ is real for all integers m, n , and

$$\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} g(m, n) \quad \text{and} \quad \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} g(m, n)$$

are both defined, then they are equal.

4. If the real valued functions $h_n(x)$ are continuous on the unit interval, and $\lim_{n \rightarrow \infty} h_n(x) = h(x)$ for all x , then $h(x)$ is a continuous function of x .